

Biomass production and carbon balance in two hybrid poplar (*Populus euramericana*) plantations raised with and without agriculture in southern France

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Abstract Poplar is useful in different climates for bioenergy production and carbon sequestration when planted as a single species or in agroforestry. Europe has large areas potentially suitable for poplar forestry and a bioenergy policy that would encourage poplar forestry. In this study I estimated biomass production and carbon sequestration in poplar monoculture plantation and poplar-wheat agroforestry, in the Mediterranean region of France. A single-tree harvesting method was used to estimate biomass and an empirical conversion factor was adopted to calculate sequestered carbon. Total biomass was higher in agroforestry trees (1223 kg tree⁻¹) than in monoculture plantation trees (1102 kg tree⁻¹). Aboveground and belowground biomass distributions were similar in both cases (89 and 88% aboveground, and 11 and 12% belowground, respectively in agroforestry and monoculture). The partitioning of total biomass in an agroforestry tree in leaves, branch, and trunk (aboveground), and fine

roots, medium roots, coarse roots and underground stem (belowground) was 1, 22, and 77, and 6, 9, 44 and 40%, respectively. Except for branch and trunk, all other compartments were similarly distributed in a monoculture tree. Storage of C was higher in agroforestry trees (612 kg tree⁻¹) than in monoculture trees (512 kg tree⁻¹). In contrast, C storage on a per hectare basis was lower in agroforestry (85 Mg ha⁻¹) than in monoculture (105 Mg ha⁻¹) due to the lower density of trees per hectare in agroforestry (139 trees in agroforestry vs 204 trees in monoculture). On a per hectare basis, soil C stocks pattern were similar to per tree stocking: They were higher in agroforestry at 330 Mg ha⁻¹ than in monoculture 304 Mg ha⁻¹. Higher C accumulation by agroforestry has a direct management implication in the sense that expanding agroforestry into agriculture production areas with short rotation and fast growing trees like poplar would encourage quicker and greater C sequestration. This could simultaneously fulfil the requirement of bioenergy plantation in Europe.

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Introduction

Estimates of the amounts of energy and organics stored in a vegetation system as soil organic matter and biomass (Satoo 1970) are essential for determining the status and flux of biological materials and understanding carbon (C) dynamics (Anderson 1970). In a scenario of changing world climate, the storage and flux of C will remain a major global concern that can be partially addressed by decreasing fossil fuel emissions, or by recapturing C

through afforestation and replenishing soil C (Guo and Gifford 2002; Oelbermann et al. 2004; Pibumrung et al. 2008; Tobias et al. 2011). Fast growing and short rotation species like poplars are major C depositories and have great potential to offset CO₂ emission (Gucinski et al. 1995; Laclau 2003; Updegraff et al. 2004; Yemshanov and McKinney 2008; Fortier et al. 2010b). Short rotation forestry crops are increasingly important in many countries because they are used in agroforestry systems that sequester greater quantities of C (Peichl et al. 2006; Maia et al. 2007; Calfapietra et al. 2010) than do pastures (Kirby and Potvin 2007), monoculture agro-ecosystems (Oelbermann et al. 2004) or conventional agricultural practices (Quinkenstein et al. 2009). Hybrid poplar has the advantage of sequestering 4–8 times more C in intercropping systems as compared to conventional agriculture (Anon. 1991; Thevathasan and Gordon 2004). In addition, hybrid poplar can enhance restoration of forest cover, accumulation of nutrients, quality of microclimates, stability of soils, and resistance to erosion within agroecosystems in a short time period (Fortier et al. 2010b). Hybrid poplar has the added advantage of supplying pulp and paper, lumber, veneer and plywood, composite panels, structural composite lumber, pallets, furniture components, fruit baskets, containers and chop and match sticks (Balatinecz and Kretschmann 2001). On account of greater biomass, superior growth, greater adaptability, and improved form and fibre characteristics, poplar plantation could become a suitable strategy to achieve the 2050 goal of 75% of timber supply from planted forest and 50% from fast growing plantations (Ceulemens et al. 1992; Heilmen et al. 1994; Sedjo 2001; Pallardy et al. 2003; Truax et al. 2012).

Poplar agroforestry has become an appealing option for sequestering C on agricultural land because it can sequester large amounts of C while leaving the bulk of the land free for agricultural production (Kort and Turnock 1998; Oelbermann et al. 2004; Peichl et al. 2006; Schoeneberger 2009). Poplar is an important fast growing soft wood, both in subtropical and temperate regions, and is cultivated widely with major agriculture crops like wheat, chickpea, and sugarcane (Jha 1999; Dupraz and Liagre 2008). *Populus euramericana* “I-214” represents the bulk of poplar plantations (Barrio-Anta et al. 2008) in European countries. The use of biomass for energy production is an important European Union Policy which aims to mitigate the effects of climate change by reducing greenhouse gas emission and securing energy supply through the diversification of energy sources (Gasol et al. 2009; Nasso et al. 2010; Rytter 2012). At the same time, overproduction of agriculture crops, high subsidies to the agriculture sector, resource protection, and the future of farmers have been concerns in Europe for many years. Therefore, agroforestry has recently been proposed as an alternative land use system

and target regions have been identified for productive growth of trees like poplars. The potential growth area of these species has been categorised as temperate and Mediterranean Europe. The Mediterranean region, in particular southern France, has a large area with potential for agroforestry (Alías et al. 2015) and high potential for growing these species in agroforestry systems (Reisner et al. 2007) yielding high profits (Graves et al. 2007). Poplar plantation has been used on a large scale outside Europe as well to meet the hardwood pulp demand in North America (Lu et al. 2011). China is also growing poplar on a large scale to meet bioenergy demands (Fang et al. 2007; Zhang et al. 2009). In India it occupies the primary position in the agroforestry system, especially the northern region (Puri et al. 1994; Singh et al. 1997; Singh 1998).

Several studies have been carried out on poplars, but C studies especially in agroforestry in Europe are limited. Therefore, this study was aimed at estimating biomass production and C distribution in different compartments of vegetation in two differently cultivated poplar systems, monoculture (MPS) and agroforestry (AFS), in the French Mediterranean. Another objective was to examine the utility of these two systems from the viewpoint of C sequestration potential.

Materials and methods

Study site

The two plantations of *Populus euramericana* clones I-214 established in 1996, (1) poplar-wheat inter-plantation (wheat-poplar agroforestry system) oriented north–south and (2) poplar monoculture plantation (monoculture poplar system) were located side by side in Vezénobres (4°9'E, 44°2'N, elevation 138 masl), southern France (Fig. 1). AFS trees were separated by 16 m between rows and 4.5 m between trees while trees in MPS were spaced 7 m between rows and trees. Trees were pruned up to 6 and 10 m following a block design. The agriculture crop was durum wheat with fallow periods every 3 or 4 years. For the last 3 years in 2009, the year of study, AFS was devoid of agriculture. Simplified soil preparation was done by disking to 10 cm depth to prepare the seed bed for the intercrop (Mulia 2005). The present study was focused on 6 m pruned I-214 clone for biomass/carbon estimation in both plantations. The climate of the study area is sub-humid Mediterranean with an average temperature of 14.8 °C and average annual rainfall of 1172 mm (years 1996–2003, Mulia and Dupraz 2006). The soil is a sandy alluvial fluvisol with 8% clay, 42% silt, and 50% sand, but pure sand and gravel layers occur at various depths in the profile, at about 1.1–1.3 and 2.5–2.9 m soil depth.

Fig. 1 Location of study site, Vezénobre in France, after harvesting the poplar plantations in 2015. Aerial photo at the bottom, taken in 2011 shows the position of two plantations (lower block in the right side): wider rows on the left is poplar-wheat inter-plantation (AFS) and narrower rows on the right of AFS is poplar monoculture plantation (MPS) (Map and photo courtesy: INRA, Montpellier and Google Earth)



Biomass estimation

The tree harvesting and dry matter estimation method which is more accurate but time and resource consuming was chosen for the present study. As a trade-off strategy, single tree harvesting (Fang et al. 1999) was done in both AFS and MPS during summer 2009. Although not having replication of harvested tree may be considered a disadvantage, sufficient care has been taken in tree selection. Tree selection was done on following parameters: (1) it was representative of the plantation having average diameter at breast height (dbh) of all the trees in the plantation, (2) it was from inner area of the plantation not the border, (3) its neighbouring trees had normal form and vigour, and (4) AFS and MPS trees both were of same clone (I-214) and same silvicultural treatments (6 m pruning). Selected AFS tree matched all these qualifications *in toto*. But MPS tree was of little higher girth (141 cm) than average (136 cm) of the plantation. Therefore, volume and biomass calculations were normalized by a factor 0.93 (square of the ratio of average tree and harvested tree) in this case.

Above ground biomass

The selected tree was felled, length (height) and girth at breast height (1.3 m) were measured, branches were numbered and trunk was cut into 1 m logs. Mid girth of these logs was recorded and then sample disc was sawn off. After measuring the diameter, long, medium and small size branches were selected in the lower, middle and upper canopy. Long and short shoot leaves from all these nine branches were collected and weighed before the branches were cut into pieces and weighed. All woody and non-woody samples were dried until constant weight in the oven at a temperature of 60 °C to remove moisture. Afterwards, the volume of trunk discs from different heights was measured by water displacement method (Nogueira et al. 2007; Basuki et al. 2009; Paladinic et al. 2009).

The trunk biomass was calculated by using the following equations:

$$B = \sum_{i=1}^n V * D \quad (1)$$

$$V = \sum_{i=1}^n S * L \quad (2)$$

$$D = \sum_{i=1}^n W/V \quad (3)$$

where, B = trunk biomass, V = volume, D = density, S = cross section area, L = length, W = weight, $n = 46$ in AFS and 45 in MPS (Ahmad and Nizami 2015; Ostadhashemi et al. 2014).

The branch and leaf biomass was calculated by using a regression equation developed between the girth and dry weight of the sampled branch and between branch girth and leaf biomass, respectively.

Below ground biomass

The soil excavation method was used for collecting coarse root (> 10 mm), medium sized root (2–10 mm) and fine root (< 2 mm) from one quarter of the rooting space. The soil pick (MBW, Slinger, WI, USA) method was used to dig the voxels (volume elements of soil analogous to pixels of 1 m × 1 m × 0.5 m size) and collect the roots. Stump root was also excavated and cut into pieces, voxel wise. Then the primary roots were shaved off the stump root of different voxels. The stool (above ground part of the stump), voxel-stump, and voxel-roots thus recovered were weighed on a field balance. Samples of these components were also weighed. Voxel samples of root were manually cleaned and categorized as above. Afterwards properly washed subsamples were drawn and dried at 60 °C temperature till constant weight. The root biomass was calculated using green and dry weight ratio.

Sequestered carbon estimation

As reviewed in Jha (2015), the estimates of biomass are essential for estimating productivity and C fluxes, which provide means of assessing C sequestration in wood, leaves and roots (Cooper 1983; Chambers et al. 2001; Specht and West 2003). Therefore, carbon stock estimation was done by using a global conversion factor 0.5 (Lamlom and Savidge 2006; Takimoto et al. 2008; Basuki et al. 2009) to biomass (Gower et al. 2001; Nowak and Crane 2002; Terakunpisut et al. 2007).

Soil carbon estimation

For the calculation of soil C storage existing data of carbon concentration and bulk density (procured from Metral and Adrianarisoa, respectively as Pers. com.) were used in the following formula.

$$SC = Cc * Bd * Sv$$

where, SC = Soil carbon, Cc = Carbon concentration, Bd = Bulk density, Sv = rooted Soil volume.

Results and discussion

Assessment of different parameters related to sampled AFS and MPS trees are recorded in Table 1. These two trees had almost similar dendrometric measurements; while the height was the same (30.7 m) the girths were 1.39 and 1.41 m, respectively of AFS and MPS tree. However, biomass controlling parameters like volume and wood density differed considerably. AFS tree produced 13% more wood volume and 10% more biomass than MPS.

Table 1 Synthesis of the results obtained from sampled tree of poplar-wheat inter-plantation (AFS) and poplar monoculture plantation (MPS). All the weights are the dry mass

Parameters	Unit	Plantations	
		AFS	MPS
<i>Growth</i>			
Girth at breast height	m	1.39	1.41
Height	m	30.7	30.7
<i>Bio-volume</i>			
Branch	m ³	0.62	0.48
Trunk	m ³	2.20	2.00
Total	m ³	2.82	2.48
<i>Biomass</i>			
Leaves	kg	13	11
Branch	kg	237	184
Trunk	kg	840	776
Above ground	kg	1091	972
Fine roots	kg	5.4	4.9
Medium roots	kg	12.4	11.6
Coarse roots	kg	58.7	58.4
Stump root	kg	53.2	54.1
Below ground	kg	132	130
Total per tree	kg	1223	1102
Total per hectare	Mg ha ⁻¹	170	209 ^a
<i>Others</i>			
Plantation density	Tree ha ⁻¹	139	204
Wood density	g cm ⁻³	0.382	0.387

^aSample tree extrapolated and normalized using correction factor

Volume and wood density

The total woody volume of AFS and MPS trees, 2.82 and 2.48 m³, respectively was apportioned in branch and trunk as 22 and 78%, and 19 and 81%. However, per hectare volume production in MPS (470 m³ ha⁻¹) was higher than AFS (389 m³ ha⁻¹). A higher stand volume production on a per hectare basis in MPS than AFS was primarily due to the higher density of plantation in the former. This was consistent with some earlier works reporting that higher stocking produced more volume of wood (Sobachkin et al. 2005; Mallik et al. 2008). Pingale et al. (2014) have also confirmed that due to fast growth and better silvicultural practices, agrisilviculture ($\equiv$ AFS) has an edge over natural plantation ($\equiv$ MPS) in production.

A higher volume per tree in AFS (2.82 m³) than MPS (2.48 m³) was not expected from the AFS tree of the same height and a lower girth. This implied that the conicity of the two trees were different and the AFS tree was more cylindrical than the MPS. Arraiolos (2006) reported an almost similar volume (total wood 2.48 m³ and trunk volume 2.08 m³) from the same AFS location. However, higher tree volume on individual tree basis in AFS than MPS may be due to more growing space in the former ensuring less inter tree competition. This may also be due to cultural practices provided to AFS crop as additional treatments like, fertilizer application, longer weed free period and extra light availability. Jha and Gupta (1991) and Banerjee et al. (2009) have also suggested that growing auxiliary crops with poplar and bamboo respectively, and providing extra irrigation, fertilizer doses, weeding and hoeing during the early age of intercropping showed up to 40% increase in height and diameter—the function of volume. The woody volume apportionment in AFS tree in the present study (branch and trunk: 22 and 78%, respectively) was different from the AFS tree (branch and trunk: 16 and 84%, respectively) of Arraiolos (2006). Irrespective of such differences wood production at 13 years in the Mediterranean region, present study, was higher than the temperate region (1.46–1.51 m³ tree⁻¹) at 20 years (Graves et al. 2007).

Trunk wood density varied from bottom to top in both AFS and MPS but there was no evident pattern of increase or decrease with increasing height of the trunk. Average trunk wood density was found to be 0.382 ($\pm$ 0.056; $n = 27$) g cm⁻³ and 0.387 ($\pm$ 0.029; $n = 13$) g cm⁻³ in the AFS and MPS trees, respectively. Low wood density of present study fell in the global range of soft and hard wood Amazonian trees (0.265–0.825) reported by Nogueira et al. (2007), and was in agreement with the observation of Balatincez and Kretschmann (2001) that all poplar species has relatively low density (0.31–0.37) and a diffuse porous structure. The difference in density of AFS and MPS was

not significant but this could be due to the stocking difference in these two plantations as suggested by Kang et al. (2004).

Tree and stand biomass

While assessing the branch and foliage biomass, polynomial regression equations ($y = ax^2 + bx + c$) developed between branch girth and branch biomass (r^2 value 0.981** in AFS and 0.982** in MPS), were highly significant while between branch girth and leaf biomass only AFS ($r^2 = 0.809$ **) was highly significant but MPS ($r^2 = 0.531$ ns) was not significant (Figs. 2, 3).

Tree biomass was higher in AFS (1223 kg tree⁻¹) than MPS (1102 kg tree⁻¹) but its distribution aboveground and belowground was almost similar (89 and 88% as aboveground and 11% and 12% below ground, respectively in AFS and MPS). Tree biomass distribution in different compartments of AFS and MPS trees is presented in Fig. 4 which indicated minor variation in different components. However, the partitioning of total biomass in AFS tree in different compartments above ground: leaves, branch, trunk, and below ground: fine roots, medium roots, coarse roots and underground stem was 1, 22, 77, 6, 9, 44 and 40%, respectively. Except for branch and trunk all other compartments had almost similar contribution in MPS (1, 19, 80, 5, 9, 45, and 41%) as compared to AFS. The data indicated that leaves and fine roots contributed minimum biomass while trunk and coarse roots stored maximum biomass aboveground and underground, respectively. However, the woody compartment (trunk and branch) stored maximum biomass in both the trees (88% in AFS and 87% in MPS, respectively).

Biomass production of poplar plantation in different countries and at different age is presented in the Table S1. The wide range of above ground biomass allocation (25–171.5 Mg ha⁻¹) in Poplar in subtropical to temperate regions suggested that biomass production is dependent on climate, stocking, age and species of poplar. Above ground biomass (152–184 Mg ha⁻¹) of the present study from Mediterranean climate with 139–204 trees ha⁻¹ stocking at the age of 13 years was closer to the higher side of the global range. However, annual production (11 Mg ha⁻¹ a⁻¹ in AFS and 14 Mg ha⁻¹ a⁻¹ in MPS) was similar to the reported range of 10–15 Mg ha⁻¹ a⁻¹ in Poplar (Quinkenstein et al. 2009).

Higher biomass storage in AFS than MPS on individual tree basis but lower storage in AFS than MPS on hectare basis had the same reason—stocking as discussed for the volume. Although Strong and Hansen (1993) suggested major role of crown closure than spacing in 16 years' hybrid poplar, there were reports of spacing effect on biomass production in other species, for example, Hegazy

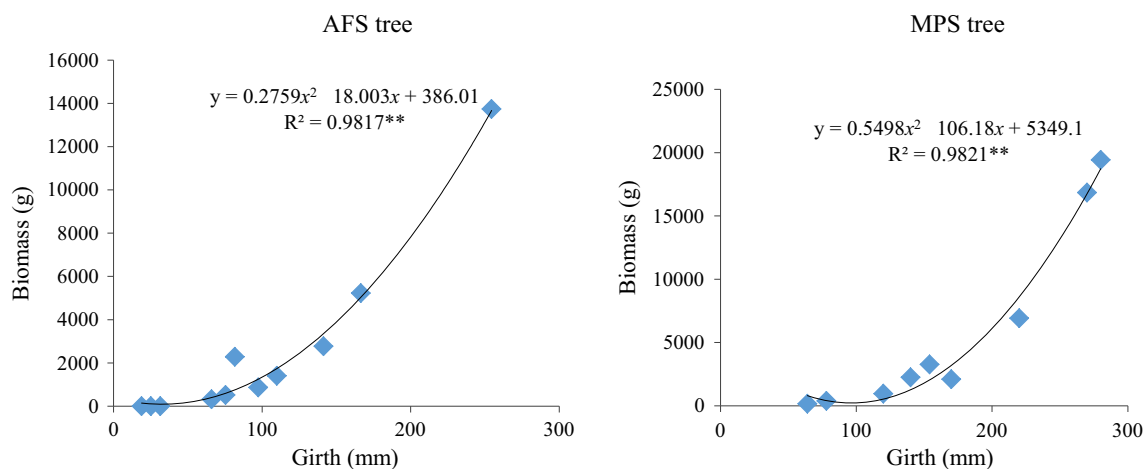


Fig. 2 Regression relation ($y = ax^2 + bx + c$) of branch biomass (y) and branch girth (x) in Agroforestry system (AFS) and Sole poplar system (MPS)

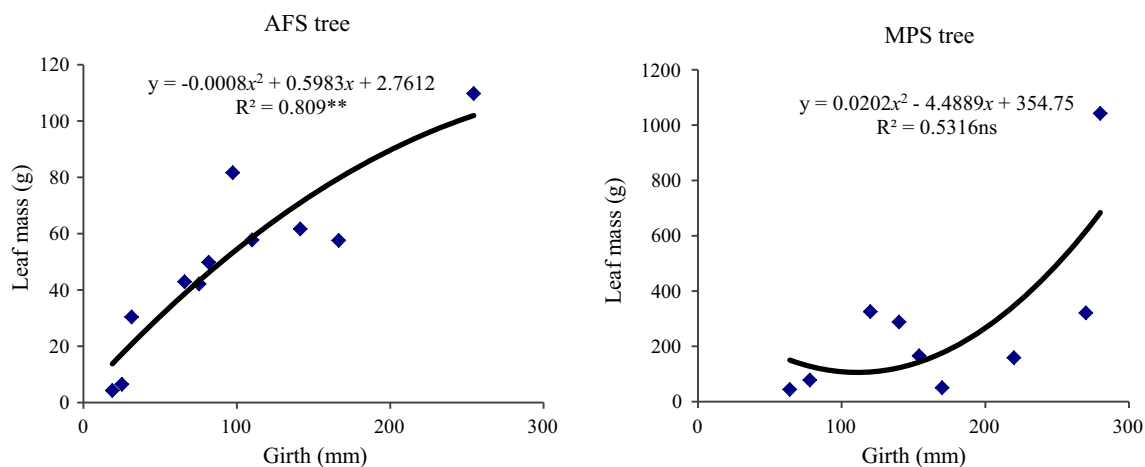
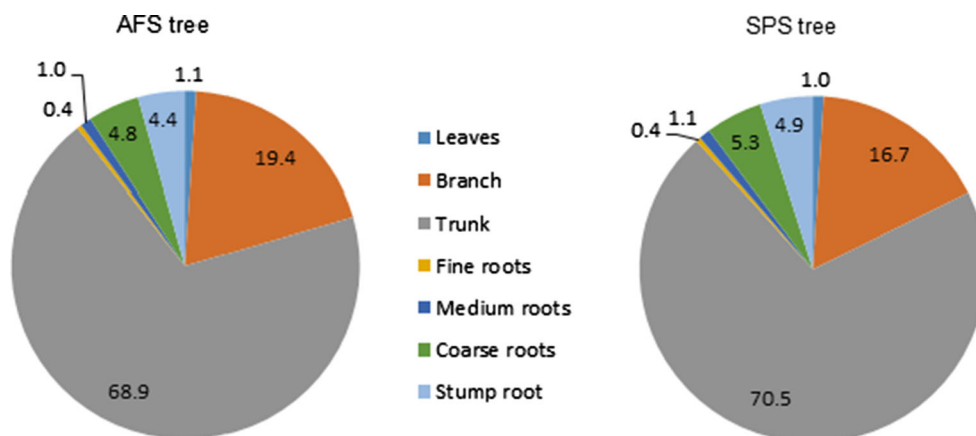


Fig. 3 Regression relation ($y = ax^2 + bx + c$) of leaf biomass (y) and branch girth (x) in poplar-wheat inter-plantation (AFS) and poplar monoculture plantation (MPS)

Fig. 4 Percentage contribution of tree components to total biomass in AFS and MPS (= SPS, Sole Poplar System) trees



et al. (2008)—higher biomass accumulation in close spacing in *Conocarpus erectus* in very low rainfall conditions and Kunhamu et al. (2009)—higher C sequestration

on hectareage basis, a function of biomass accumulation, in closed spaced *Acacia mangium* in tropical climate.

The ratio between root and shoot in the present study (11.8–13.2%) was outside the reported range of 14–35% from different poplar clones and species at different age (Supplementary Materials: Table S1). This could possibly be the result of interaction of I-214 clone of *Populus euramericana* with particular edapho-climatic conditions at the present study site, fluvisol characterised by seasonally receding water table and high evapo-transpiration potential. This was in agreement with the argument of Scarascia-Mugnozza et al. (1997) that the proportion of root: shoot allocation changes with the quality of the site along with genotype and age of the tree.

Soil carbon storage

Carbon concentration (4.03–16.54 g kg⁻¹ in 0–15 cm to 120–240 cm layer, Metral Pers. com.) and bulk density (1.1 to 1.67 in 0–15 to 250–300 cm depth, Adrianarisoa Pers. com.) from the same area of the present study were not homogeneous along the soil depth. Therefore, the average of heterogeneous data was used for C storage calculation. For 3 m depth, soil C pool per tree was higher in AFS (2.4 Mg) than MPS (1.5 Mg). On a per hectare basis the pattern was same as well. It was 330 Mg ha⁻¹ in AFS and 304 Mg ha⁻¹ in MPS. Stand storage also followed a similar pattern (AFS 415 Mg ha⁻¹ and MPS 409 Mg ha⁻¹), although the difference was marginal.

Soil C cycling, an important component of the C balance in ecosystems (Nakane 1995) is derived mainly from leaf litter and root decomposition. The lack of baseline or control data in the soil at the beginning of plantation made it difficult to quantify C sequestration/storage by MPS or AFS. However, AFS had higher C (330 Mg ha⁻¹) than MPS (304 Mg ha⁻¹) but there was not enough data to test the significance of such difference. Nevertheless, the difference between AFS and MPS gave a tentative indication that AFS accumulated more soil C (26 Mg ha⁻¹) than MPS in 13 years. This indicative result was different from an earlier report in the tropical region by Swamy and Puri (2005) who worked on *Gmelina arborea* and suggested that agrisilviculture system is inferior to plantation system. Nonetheless, recent studies in the tropical region recorded that silvopasture, another form of agroforestry, was better than other land uses (Maia et al. 2007). Further, improved agroforestry practices had an advantage over the traditional parkland system (Takimoto et al. 2009) as far as C sequestration was concerned. These two findings supported the indicative results of the present study. A recent publication on poplar agroforestry system (Gupta et al. 2009) reported that soil organic C increased with age to the tune of 1.95–2.63 Mg ha⁻¹ a⁻¹ until 6 years in the tropical domain. This is similar to the present finding of 2 Mg ha⁻¹ a⁻¹ at 13 years. However, this finding is much

lower than the latest report of 4 Mg C ha⁻¹ a⁻¹ in small holder farms and agroforestry systems in the Philippines (Lasco and Pulhin 2009). Low soil carbon in the present study could be related to the fact that the soil was sandy and heterogenous, resulting in very slow humification and in turn C accumulation.

Populus euramericana I-214 soil C pool, present study, was much higher than another hybrid poplar (*Populus deltoides* x *nigra* DN-177) of the same age in the temperate zone and traditional agroforest (78.5, 58.5, and 145 Mg ha⁻¹) reported by a few workers (Oelbermann et al. 2006; Peichl et al. 2006; Kirby and Potvin 2007). When C stocking in vegetation and its storage in soil was combined, the pattern did not change and AFS (415 Mg ha⁻¹) remained higher than MPS (409 Mg ha⁻¹) but this time the margin of difference narrowed down considerably. However, present assessment of C was much higher than the earlier report of Arevalo et al. (2009) in 9 years old Walker hybrid poplar plantation (174 Mg ha⁻¹) of the temperate zone.

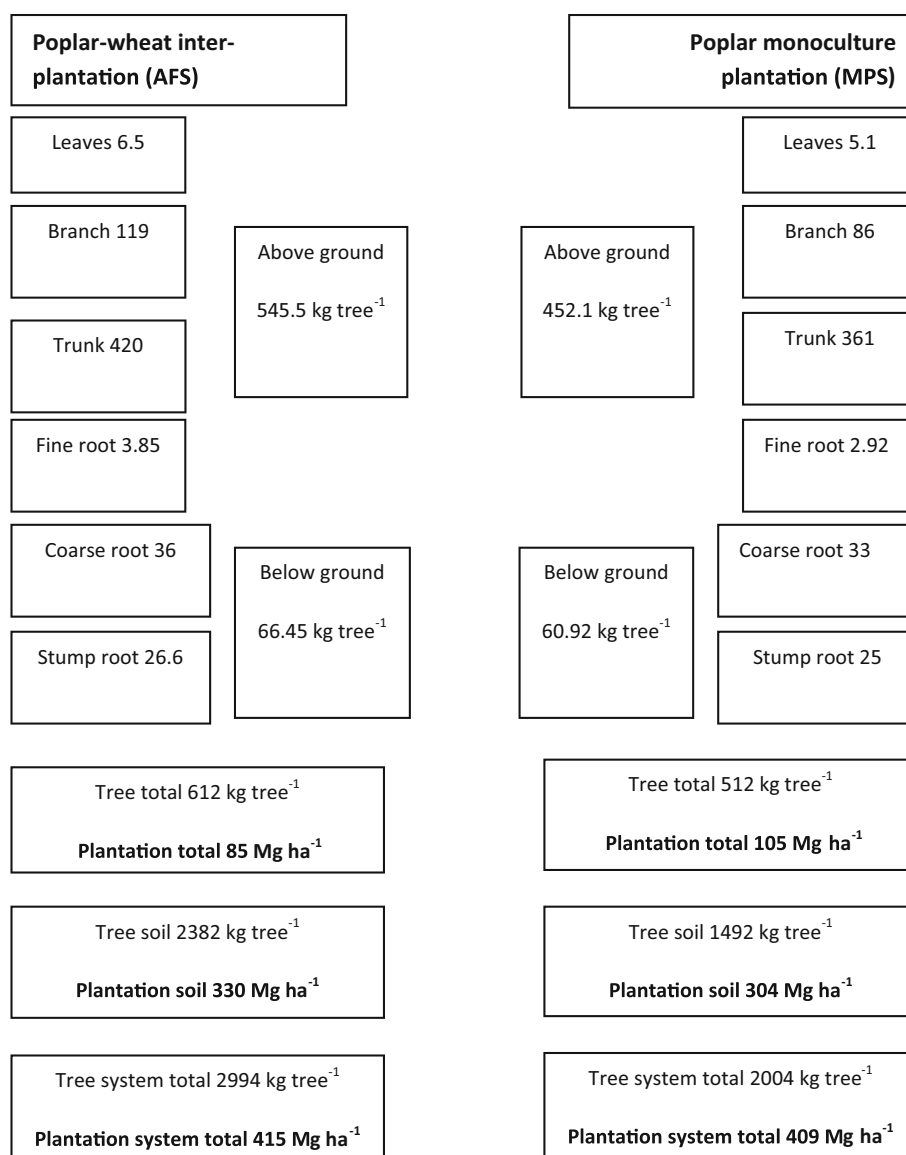
One of the sources of soil C is annual fine root decomposition. Therefore, information on fine root biomass and its production is critical since it plays an important role in the cycling of C in a system (Chen et al. 2004; Graefe et al. 2008). In general poplar fine roots have a life span of 30–365 days and in particular hybrid poplars have a span of 36–100 days (Block et al. 2006), therefore C turnover depends on its longevity. Although fine root turn over in *Populus euramericana* I-214 depends on the seasons of the year (Mulia and Dupraz 2006), the present study conducted during the summer indicated a significant amount of C contribution by this deciduous organ of the tree. Predicted value (0.85 Mg ha⁻¹ a⁻¹, unpublished data) fell in the range of 0.2–1.6 Mg ha⁻¹ a⁻¹ recorded from various studies (Block et al. 2006). Further study in terms of fine root longevity in the present study site will give an insight to accuracy of C injection through this compartment.

System carbon balance

The standing state of C presented in Fig. 5 indicated that per tree storage of C was higher in AFS (612 kg) while, per hectare storage of C was higher in MPS (105 Mg ha⁻¹). However, C stocking in soil on per hectare basis was higher in AFS (330 Mg ha⁻¹). When C storage in vegetation and its stocking in soil were combined, the pattern remained same—higher storage in AFS (415 Mg ha⁻¹) than MPS (409 Mg ha⁻¹) but with narrowed margin of difference.

An intensive search of literature suggested that there are very few reports on C balance in poplar plantations (Fang et al. 2007; Sartori et al. 2007 etc.). Therefore, going through biomass data (Table S1) and converting the

Fig. 5 Carbon balance in different components of a tree (kg tree^{-1}) and in plantation stand (Mg ha^{-1}) of two different systems—inter-plantation and monoculture—*Populus euramericana* (I-214) at the age of 13 years in Mediterranean region



biomass into half as C content revealed that its storage in above ground parts varied from 13 to 147 Mg ha^{-1} for a density range 400–8333 trees ha^{-1} and age range 4–52 years. Aboveground C assessment in AFS (76 Mg ha^{-1}) and MPS (92 Mg ha^{-1}) at 13 years fell in this range. A similar range of C accumulation (76.6–80.1 Mg ha^{-1}) in the temperate region at 22 year age (Rytter 2012) indicated faster C sinking in Mediterranean poplar plantation. But it was much slower than tropical Indian agrisilviculture plantation (66–83 Mg ha^{-1} at 7 years; Rizvi et al. 2011). However, present production was also covered by the reported range of 21.7–151.3 Mg ha^{-1} in various species, age and management specific agroforestry systems in tropical climate (Nair et al. 2009). Taking root: shoot ratio into account (Table S1) it was evident that C storage in roots was

14–35% of total C storage in plants. The present value of C storage in below ground parts does not match this range. Very low below ground storage indicated that *Populus euramericana* clone I-214 at the age of 13 years needed to invest less C in roots.

Soil C to biomass C ratio estimated in the present study (2.9–3.9) fell in the reported range of 1 (temperate forest) to 5 (boreal forests) (Arora et al. 2014; IPCC 2000). MPS (2.9) value is well within the range of terrestrial ecosystem (Post et al. 1990) and AFS (3.9) value is higher than this. Higher ratio of AFS than MPS is indicative of better C storage efficiency of the Agroforestry system. Otherwise too, excess C accumulation in AFS soil (2 $\text{Mg ha}^{-1} \text{a}^{-1}$) or overall system (0.5 $\text{Mg ha}^{-1} \text{a}^{-1}$) with reference to MPS (indicative difference) suggested that poplar agroforestry could be expanded in surplus agriculture production areas

in order to achieve the goal of C sequestration. Although the C sequestration by agriculture crops was not studied in the present case, the universal assumption is that in the longer term any traditional agricultural technique does not lead to C storage. Moreover, agroecosystem has the potential to sequester as well as emit CO₂ and agroforestry, among others, is one of the strategies to address the challenges of carbon sequestration in agricultural soil (Lal 2011; Wang et al. 2015).

System efficiency in carbon storage

The efficiency of the agroforestry system in C storing is high as compared to an agriculture system. Even if we neglect the aboveground tree pools (trunk and branches) that may not store C in the long term, the importance of below-ground C pools (coarse roots, fine roots) and of the leaf litter should be taken into consideration. However, the fate of the leaf litter is not easy to predict and measurements of soil C in the topsoil did not evidence any change in soil organic matter (SOM) during the last 12 years (Dupraz, Pers. com.). This indicated that the key component for long term C storage in the soil may be the tree roots. They are diluted in a huge volume of soil, making the direct measurement of a soil C change very difficult. It was witnessed in this work that the fine roots production represented about 0.7 Mg C ha⁻¹ a⁻¹, and that the coarse roots represented 1.4 Mg C ha⁻¹ a⁻¹. This was a total of 2 Mg C ha⁻¹ a⁻¹ that may be stored for the long term in the soil, including at deep soil depth where mineralisation may be very slow (Fontaine et al. 2007). There was, however, a doubt about the storage of the C as fresh C injection in soils may also stimulate the activity of microbes responsible for the mineralisation (Fontaine et al. 2004), resulting in a loss of mature SOM. This is an open question that should be addressed in the near future. A different hypothesis may also be structured about the interaction between the water-table and the fate of the dead (fine and coarse) roots: in saturated horizons, the organic matter decomposition is quite slow, resulting in better long term C storage. The fact that AFS trees have deeper roots as compared to MPS trees may therefore enhance the soil C sequestration by this mechanism.

Conclusion

Single tree harvesting may be contemplated as the limitation by the school which does not consider the methodology adopted in the present study precise enough but considering the trade off and modification in the single tree harvesting method (average tree selection/normalising) the data generated is dependable and can be taken for future

use. However, it has been advocated that most annual cropping systems are associated with a decline in soil organic carbon that need to be compensated by perennial bioenergy production (Don et al. 2012). As suggested by Rytter (2012) poplar plantations on abandoned arable land has the potential of increase in the rate of C sequestration and thereby mitigate the negative effects of increasing atmospheric CO₂ concentrations. Recently, it has been suggested that the future of agroforestry in Europe will become increasingly important due to woody perennials producing bioenergy and food/fodder in the same area (Nerlich et al. 2013) as along with efficient C sequestration (Sierra et al. 2013). Field measurements of biomass or C pools in AFS and MPS provided estimates of the above-ground/below-ground ratio in these two systems. Carbon storage by inter-plantation (AFS) was found higher on a per tree basis as compared to monoculture plantation (MPS) tree as a consequence of the cylindricity of the stem and a slightly higher root pool. The root biomass plays a significant role in long term C storage. Sharma et al. (2015) have also observed that Poplar based agroforestry system has the potential to make soil a net sink for C thereby attenuating CO₂ load in the atmosphere and improving soil fertility and productivity. Therefore, it may be speculated that expanding agroforestry into agriculture production areas with short rotation and fast growing trees like poplars would be judicious from a C sequestration point of view. The inter-plantation system can secure sufficient tree related benefits with some compromise on agriculture production as compared to block plantation. Chauhan et al. (2011) have also advocated agroforestry (poplar-wheat) as a better option than the sole agricultural crop not only from the CO₂ mitigation angle but also for sustainable productivity or profitability, provided the system continues for several rotations.

Although hedgerow management of tree crops is more effective in farmland biodiversity conservation (Dainese et al. 2015) and similar in soil organic matter to forests (Sitzia et al. 2014), there is a caveat that demographic swamping and genetic assimilation of hybrids with indigenous poplar may cause anxiety towards possible reduction in genetic variability of latter (Cagelli and Lefèvre 1995; Levin et al. 1996) or genetic erosion and extinction of native poplar trees, in particular *Populus nigra* (Winfield et al. 1998), which could maintain niches inside semi-natural forests and hedgerows and compete with cultivated hybrid poplar (Smulders et al. 2008; Sitzia et al. 2013). However, Fossati et al. (2003) have indicated that introductions of genetically modified poplar hybrids may not have impact on the natural poplar population. Although AFS and MPS have their own advantages, it is advisable in broader sense that a mosaic of different land uses, including agroforestry, croplands, and tree crop on

abandoned fields could be the better solution for overall benefit.

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